

Fabric Synthesizer - An analog synthesizer played by manipulating a conductive fabric.

Sponsor: Student self-direct, no sponsor

Background:

Traditional electronic synthesizers typically rely on fixed axis inputs. Some examples include a knob turning, a key or button being pressed, or a switch being flipped. While this type of input can produce a range of unique sounds and patterns, it can take a lot of patience and knowledge of the specific instrument. To reduce the complexity of playing these instruments while making it easier to produce experimental sounds, a conductive fabric interface can be used. As the fabric is manipulated by a musician, its electrical properties like resistance and capacitance will change. These changes can be used to control various parameters such as filters, effects, and oscillators.

Project Scope / Needs:

1. Conductive Fabric
 - a. Durable base material that can be manipulated easily
 - b. Flexible conductive thread to be woven in base fabric
2. Synthesizer
 - a. Senses changes in parameters of fabric
 - b. Includes filters, oscillators, effects
 - c. Includes interface to control volume
 - d. Compatible with other modular synthesizers
3. MIDI Input
 - a. Allows synthesizer parameters like pitch to be controlled by an external keyboard or computer
4. Output
 - a. Internal speaker

Technical Considerations:

1. Conductive Fabric
 - a. Design layout for conductive thread and chose materials
2. Synthesizer
 - a. Implement sensors to detect changes in resistance and capacitance of fabric
 - b. Generate basic wave shapes using voltage controlled oscillators
 - c. Design audio effects like delay, distortion, and reverb
 - d. Design power supply
 - e. Fabricate a simple enclosure for circuitry
 - f. Safely interact with other modular synthesizers
3. MIDI Input
 - a. Program microcontroller to work with any external MIDI controller
 - b. Design a DAC to control pitch of synthesizer
4. Speaker
 - a. Design an amplifier with volume dial to power a built in speaker

Deliverables:

1. Functional prototype synthesizer with MIDI compatibility, fabric interface, and adjustable speaker.
2. Documentation of design decisions.